

Adem Yahmadi

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PROFILE

Computer Science student at TUM interested in building software for real-world systems. Experience in backend development, numerical optimization, and applied AI, with projects spanning autonomous mobility, infrastructure monitoring, and computer vision.

EDUCATION

- **Technical University of Munich** 2023 – Present
Bachelor of Science in Informatics Munich, Germany
- **Lycée Hrairia 2** 2019 – 2023
Scientific Baccalaureate, Mathematics (Highest Honors) Tunis, Tunisia

EXPERIENCE

- **TUfast Eco Team** 2024 – 2025
Driving Strategy Engineer Technical University of Munich
 - Developed the race strategy optimizer for the lux025 solar car (Bridgestone World Solar Challenge 2025), planning speed profiles over 3,000 km under energy and race constraints.
 - Extended the PyTorch vehicle model to optimize speed profiles with gradient descent and integrated multiple optimization solvers (IPOPT, ParOpt, ALPSO).
 - Built a PySide6 GUI for running optimizations and wrote unit tests for the control-stop simulation (pytest, GitLab CI).

PROJECTS

- **InfraLens – Proxmox Cluster Visualizer & AI Assistant**  [Live](#)  [Code](#)
Python, FastAPI, React, Tauri, Proxmox API, Ollama, SSH
 - Built a Tauri/React desktop app that connects to Proxmox clusters, visualizes nodes, VMs, and containers as an interactive topology graph, and monitors resources in real time.
 - Developed a FastAPI backend for VM and container controls and SSH-based service discovery.
 - Integrated a local Ollama LLM that uses live cluster state for natural-language infrastructure diagnostics.
- **Vertex – Intelligent Document Scanner**  [Live](#)  [Code](#)
Python, FastAPI, OpenCV, React, Docker
 - Built a scanning pipeline using deep learning (DocAligner) and OpenCV for edge detection, perspective correction, and shadow removal.
 - Developed a CLI and React web app that converts photos into full-resolution PDFs; deployed with Docker on Render.
- **mdtopdf – Markdown to PDF Converter**  [Live](#)  [Code](#)
Node.js, Express, KaTeX, Mermaid, Supabase, Docker
 - Built a CLI and web app that converts Markdown to print-ready PDFs, with LaTeX math (KaTeX), Mermaid diagrams, and themed syntax highlighting for 200+ languages.
 - Developed a live-preview editor with PDF export, Supabase auth and cloud storage; deployed with Docker on Render.

UNIVERSITY PROJECTS

- **Motion Planning for Autonomous Vehicles** WS 2025/26
Python, CommonRoad, Shapely, Matplotlib
 - Converted drone-recorded traffic data into CommonRoad simulation scenarios for benchmarking autonomous vehicle motion planners.
 - Built a validation tool with spatial indexing to detect collisions, off-road driving, and physically infeasible motion, with visualizations for debugging.
- **Computer Architecture & Fundamentals of Programming** WS 2023/24 – SS 2024
C, C++, SystemC, Java
 - Built a cycle-accurate SystemC cache simulator comparing direct-mapped and associative caches.
 - Solved 35+ Java assignments on OOP, recursion, data structures, algorithms, testing, and concurrency.

SKILLS

- **Programming:** Python, Java, C++, C, JavaScript
- **Frameworks & Libraries:** FastAPI, React, Node.js, PyTorch, NumPy, OpenCV
- **Tools & Infrastructure:** Linux, Docker, Git, GitLab CI, pytest, SSH, Proxmox, Ollama
- **Languages:** German (native), English (C1), Arabic (native), French (C1)